SUPER ADMIN PLATFORM

Technical Specification & Implementation Guide — v1.0 · March 2026

System: **School Management SaaS Platform**

The **Super Admin Platform** is the operational command center of the system.
It manages tenants (schools), billing, subscriptions, system monitoring, feature control, security, compliance, and operational tooling.
This module ensures the platform can **scale safely, prevent financial losses, and maintain operational stability**.

1. MODULE OVERVIEW & SUBMODULE STRUCTURE

The Super Admin Platform contains **15 core platform submodules** responsible for infrastructure management, tenant operations, billing, monitoring, and platform governance.

Submodule	Description	Priority	Status
Platform Dashboard	System overview, metrics and operational health	Core	Pending
Tenant Management	Manage schools (tenants) and lifecycle	Core	Pending
Subscription & Billing	Plans, invoices, payments and lifecycle	Core	Pending
Usage Metering	Resource usage tracking and limits	Core	Pending
Feature Flags	Enable/disable features per tenant	Core	Pending
Monitoring & Health	System infrastructure monitoring	Core	Pending
Audit Logs	System-wide activity tracking	Core	Pending
Security & Access Control	RBAC, authentication and security policies	Core	Pending
Backup & Disaster Recovery	Database backup and restoration tools	Core	Pending
Deployment Management	Platform releases and version control	Core	Pending
Support & Impersonation	Support tools for debugging tenant issues	Core	Pending
Compliance & Data Privacy	GDPR/COPPA data governance tools	Enhanced	Pending
Notifications & Broadcast	Platform announcements and alerts	Enhanced	Pending
API & Integration Management	External integrations and API keys	Enhanced	Pending
Revenue Analytics	SaaS metrics like MRR, churn, LTV	Enhanced	Pending

2. PLATFORM DASHBOARD

Purpose

Provides a **real-time operational overview** of the platform.

Super admins can view:

- tenant statistics
- revenue
- system performance
- infrastructure health
- alerts

Key Features

System Overview

- Total number of tenants
- Active vs suspended schools
- Total users across tenants
- Platform uptime

Revenue Metrics

- Monthly recurring revenue (MRR)
- Annual recurring revenue (ARR)
- Subscription distribution by plan

System Health

- API response time
- database load
- job queue backlog
- storage usage

Alerts

- billing failures
- high system latency
- server outages
- suspicious login activity

Data Models

Platform Metrics

Field	Type	Description
id	integer	Primary key
metric_name	string	Metric identifier
metric_value	decimal	Value
recorded_at	datetime	Timestamp

API Endpoints

Method	Endpoint	Description
GET	/api/platform/dashboard/summary/	System overview
GET	/api/platform/dashboard/revenue/	Revenue metrics
GET	/api/platform/dashboard/system-health/	Infrastructure metrics

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3. TENANT MANAGEMENT

Purpose

Manages all **schools using the platform**.

Key Features

Tenant Lifecycle

- create tenant
- update tenant
- suspend tenant
- reactivate tenant
- delete tenant

Tenant Metadata

- school name
- contact information
- plan

- domain
- location

Tenant Status

- trial
- active
- suspended
- cancelled

Tenant Isolation

All tenant data must include:

```
tenant_id
```

Data Models

Tenant

Field	Type	Description
id	UUID	Tenant ID
name	string	School name
domain	string	Subdomain
plan_id	FK	Subscription plan
status	enum	Trial / Active / Suspended
created_at	datetime	Creation timestamp

API Endpoints

Method	Endpoint	Description
GET	/api/platform/tenants/	List tenants
POST	/api/platform/tenants/	Create tenant
GET	/api/platform/tenants/{id}/	Tenant details
PATCH	/api/platform/tenants/{id}/	Update tenant
POST	/api/platform/tenants/{id}/suspend/	Suspend tenant

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4. SUBSCRIPTION & BILLING

Purpose

Handles SaaS billing operations.

Key Features

Plan Management

- create pricing plans
- configure limits
- upgrade/downgrade tenants

Billing Cycle

- monthly
- yearly

Invoice Management

- generate invoices
- payment status tracking
- invoice history

Data Models

Plans

Field	Type	Description
id	UUID	Plan ID
name	string	Plan name
price	decimal	Monthly cost
student_limit	integer	Max students
storage_limit	integer	Storage quota

Subscriptions

Field	Type	Description
id	UUID	Subscription ID
tenant_id	FK	Tenant
plan_id	FK	Plan
status	enum	Active / Cancelled
next_billing_date	date	Next charge

Invoices

Field	Type	Description
id	UUID	Invoice ID
tenant_id	FK	Tenant
amount	decimal	Invoice amount
status	enum	Pending / Paid

API Endpoints

GET /api/platform/plans/
POST /api/platform/plans/
GET /api/platform/subscriptions/
POST /api/platform/invoices/

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5. USAGE METERING

Tracks tenant resource consumption.

Key Metrics

- student_count
- storage_usage
- SMS usage
- API requests

Data Models

Usage Metrics

Field	Type
tenant_id	FK
metric_name	string
metric_value	integer
period	date

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6. FEATURE FLAGS

Allows selective feature rollout.

Example Flags

- AI reports
- advanced analytics
- attendance biometrics
- LMS module

Data Model

Feature Flags

Field	Type
id	integer
name	string
description	text
enabled	boolean

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7. MONITORING & SYSTEM HEALTH

Monitors infrastructure.

Typical monitoring tools used by companies like
Amazon Web Services include platforms such as Prometheus.

Metrics

- CPU usage
- memory
- API latency
- database queries
- queue backlog

8. AUDIT LOGS

Tracks all critical actions.

Example Events

- tenant created
- billing updated
- impersonation started
- system configuration changed

Data Model

Audit Logs

Field	Type
actor_id	FK
action	string
target	string
timestamp	datetime

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9. SECURITY & ACCESS CONTROL

Implements **role-based access control (RBAC).**

Roles

Role	Access
Owner	Full access
Platform Admin	Platform operations
Billing Admin	Billing operations
Support Agent	Tenant support
Auditor	Read-only logs

Security Controls

- Two-factor authentication
- session expiration
- IP monitoring
- login anomaly detection

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10. BACKUP & DISASTER RECOVERY

Protects against data loss.

Backup Strategy

- daily database backups
- weekly snapshots
- monthly archives

Recovery Options

- full database restore
- tenant restore
- point-in-time recovery

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11. DEPLOYMENT MANAGEMENT

Handles platform updates.

Deployment Pipeline

```
Development
↓
Staging
↓
Production
```

Version control typically managed via platforms such as GitHub.

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12. SUPPORT & IMPERSONATION

Allows support team to debug tenant issues.

Features

- tenant impersonation
- session logging
- auto session expiration
- troubleshooting tools

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13. COMPLIANCE & DATA PRIVACY

Required because schools manage **children's data**.

Compliance Tools

- data retention policies
- right-to-erasure requests
- consent logs
- data export requests

Regulatory frameworks include:

- General Data Protection Regulation
- Family Educational Rights and Privacy Act

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14. NOTIFICATIONS & BROADCAST

Allows communication to tenants.

Features

- platform announcements
- maintenance alerts
- billing notifications
- email/SMS alerts

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15. API & INTEGRATION MANAGEMENT

Manages third-party integrations.

Integration Examples

- payment gateways
- messaging services
- learning platforms

API Keys

Each tenant can generate:

- API keys
- webhook endpoints

16. REVENUE ANALYTICS

Tracks SaaS business health.

Metrics

- MRR (monthly recurring revenue)
- ARR (annual recurring revenue)
- churn rate
- LTV (lifetime value)

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17. INTEGRATION POINTS

From	To	Integration
Platform	Billing	Subscription enforcement
Platform	Communication	Broadcast notifications
Platform	Finance	Revenue reporting
Platform	HR	User authentication

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18. PERMISSIONS & ACCESS CONTROL

Module	Owner	Admin	Support	Auditor
Tenant Management	R/W	R/W	R	R
Billing	R/W	R/W	R	R
Audit Logs	R	R	R	R